

MATH FOR AI • EPISODE 0.1

Why AI Is Math in Disguise

Data, models, and learning: the three ideas behind modern AI

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Face recognition. Chatbots. Self-driving cars.

Under the hood, every one of them is doing
math.

And it is friendlier math than you might think.

Three ideas behind modern AI

IDEA 1

Data

Everything becomes numbers.

IDEA 2

Model

A function turns an input into a prediction.

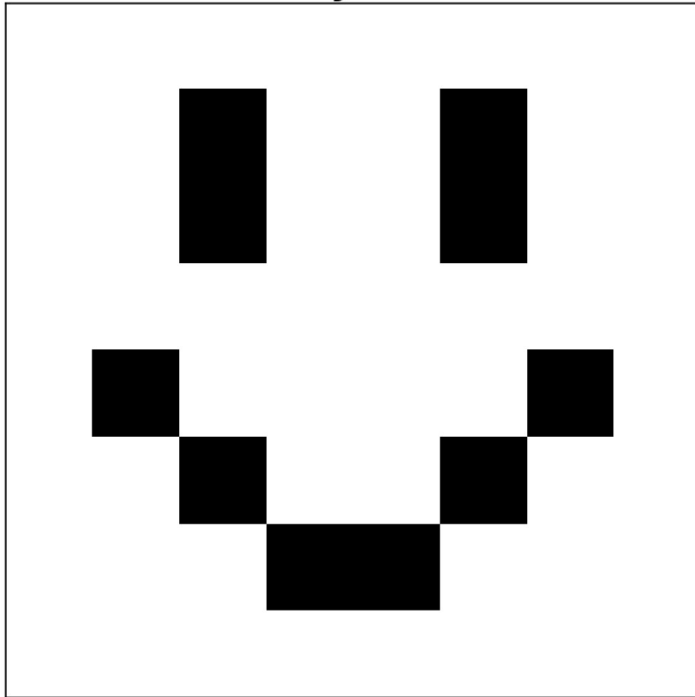
IDEA 3

Learning

Nudge the numbers until the model is less wrong.

Idea 1: Data is just numbers

What you see

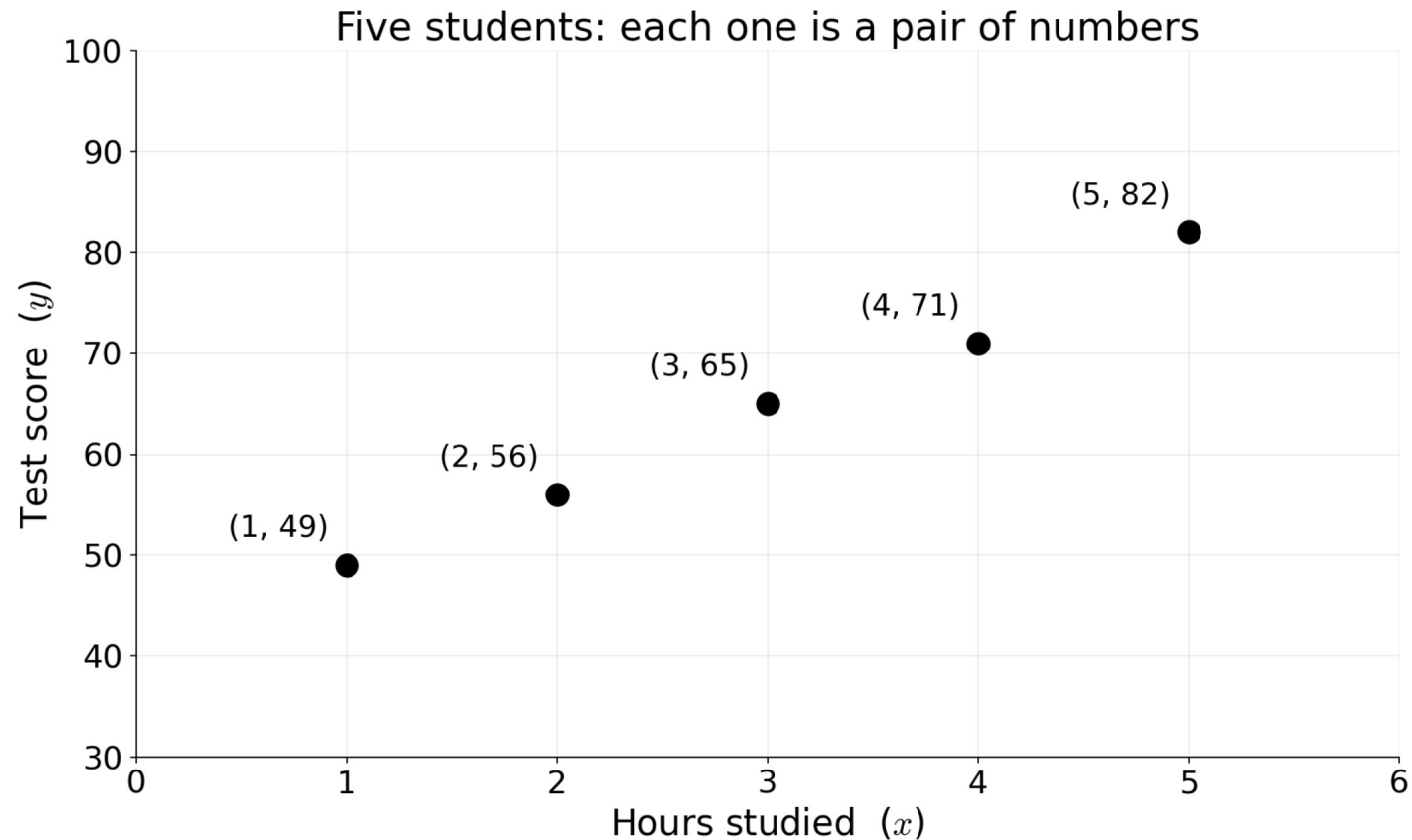


What the computer sees

9	9	9	9	9	9	9	9
9	9	0	9	9	0	9	9
9	9	0	9	9	0	9	9
9	9	9	9	9	9	9	9
9	0	9	9	9	9	0	9
9	9	0	9	9	0	9	9
9	9	9	0	0	9	9	9
9	9	9	9	9	9	9	9

Real photos: millions of cells, each with red, green and blue values from 0 to 255.

Our example: hours studied $\rightarrow$ test score



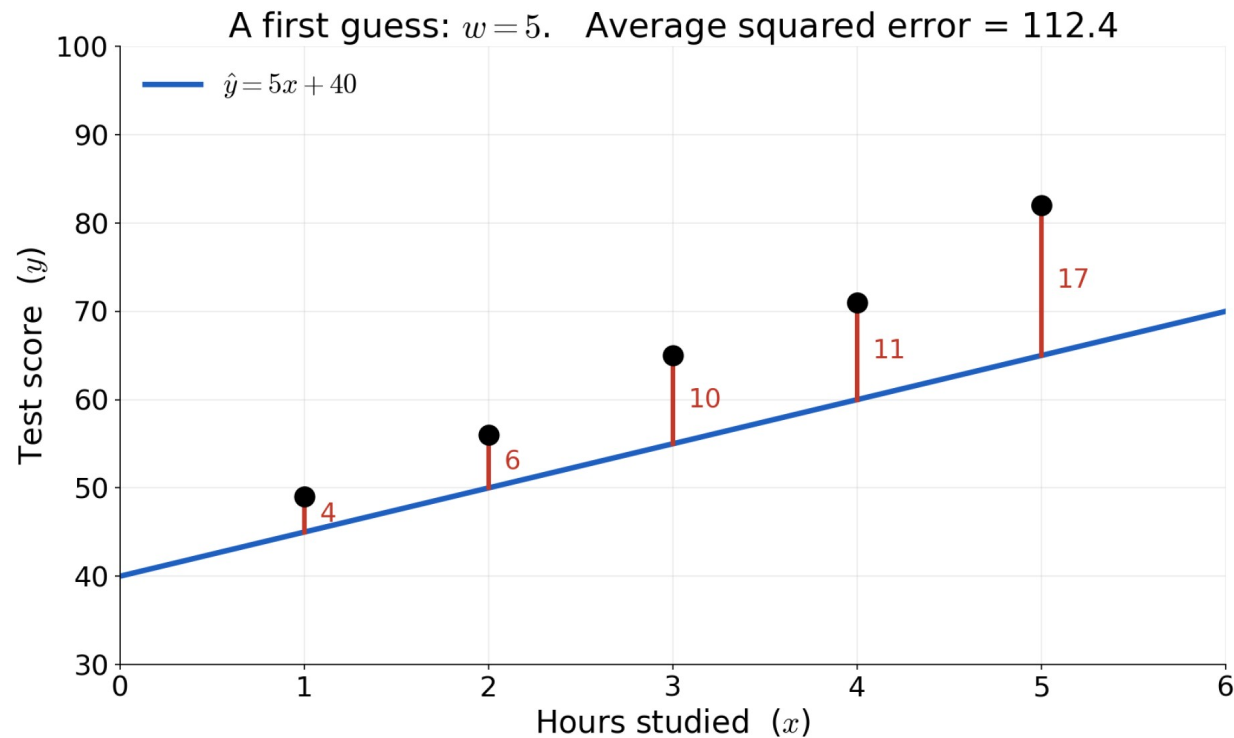
Idea 2: A model is a function

$$\hat{y} = wx + b$$

Symbol	Say it	What it means here
$\hat{y}$	“y-hat”	the prediction: our guessed score
x	“x”	the input: hours studied
w	“w” (weight)	the slope: points gained per hour
b	“b” (bias)	the starting score with zero studying

A first guess: $w = 5$

For today we keep $b = 40$ and let the computer learn only w . Red bars show each mistake.



Idea 3: Measure how wrong we are

$$\text{Error} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

1. Subtract

real score minus predicted score

2. Square

makes every miss positive and
punishes big misses

3. Average

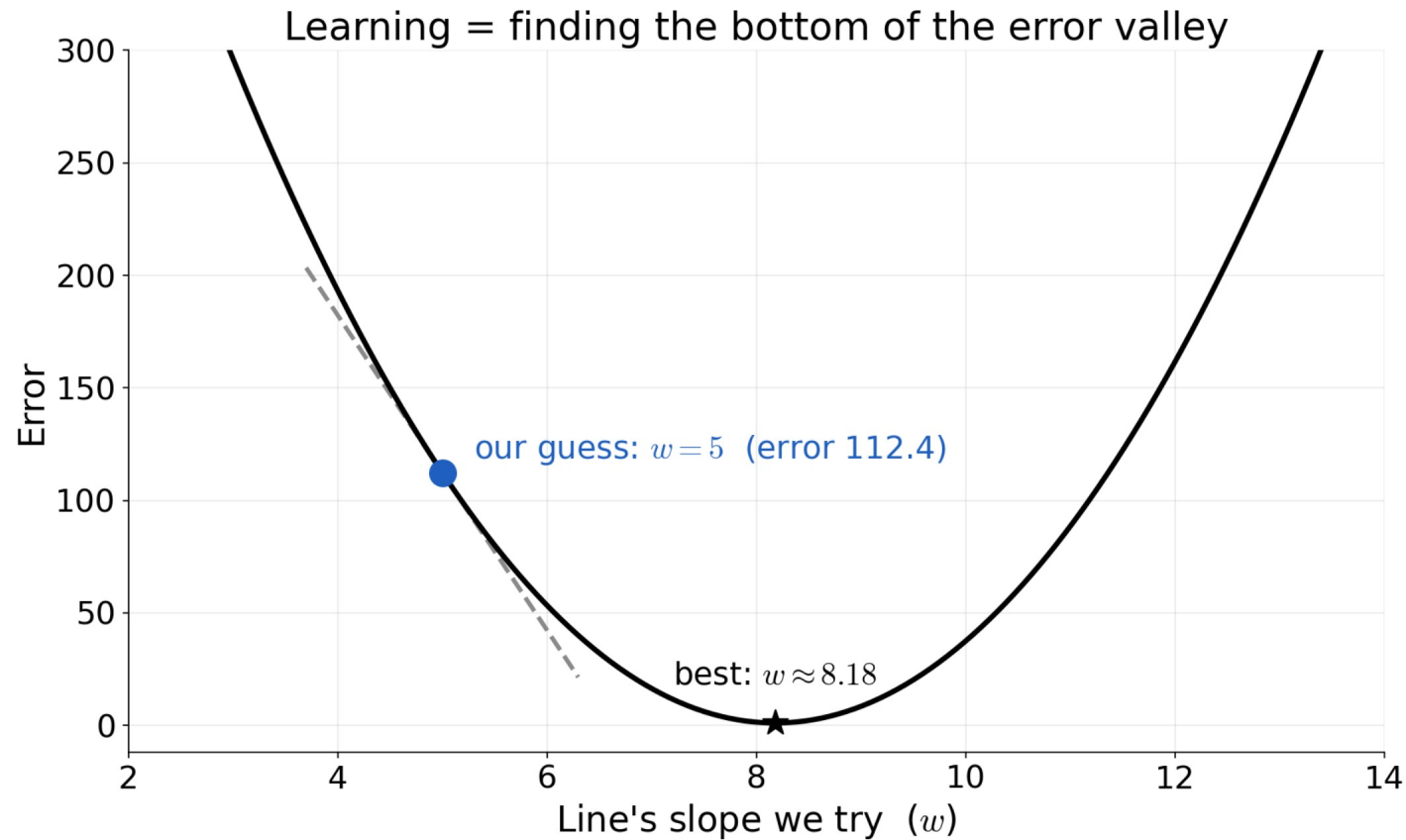
add them all up, then divide by n

Let's compute it by hand ($w = 5$)

Hours x	Score y	Guess $\hat{y} = 5x + 40$	Miss $y - \hat{y}$	Miss ²
1	49	45	4	16
2	56	50	6	36
3	65	55	10	100
4	71	60	11	121
5	82	65	17	289

Add: $16 + 36 + 100 + 121 + 289 = 562$. Average: $562 \div 5 = \mathbf{112.4}$

Every w has an error: the error valley



Which way is downhill?

At $w = 5$, the valley's **steepness** is **-70**. Negative means the error drops as w grows, so we step to the right.

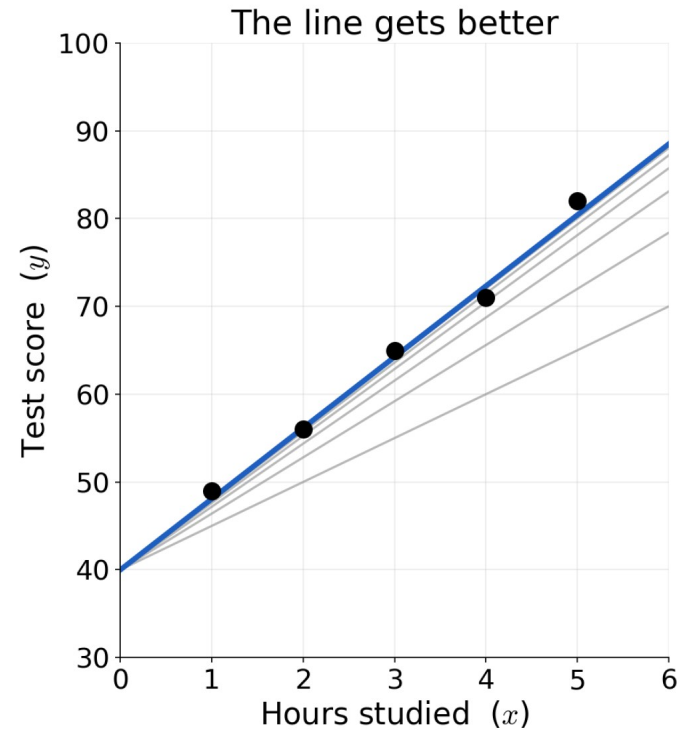
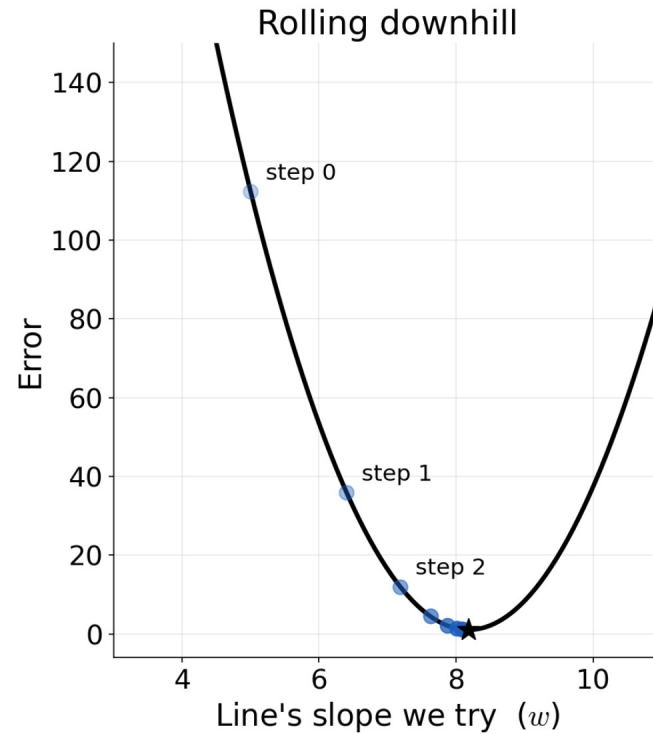
Two different things: w is the slope of our **line**. Steepness is how fast the **error** changes when we nudge w .

$$w_{\text{new}} = w - \eta \cdot \text{steepness}$$

η (“eta”) is the step size, also called the learning rate. We use $\eta = 0.02$.

$$w_{\text{new}} = 5 - 0.02 \cdot (-70) = 5 + 1.4 = 6.4$$

Repeat, and roll downhill



Error: $112.4 \rightarrow 36.0 \rightarrow 12.0 \rightarrow 4.5 \rightarrow 2.1 \rightarrow 1.4 \rightarrow 1.1$

The whole loop, from scratch in Python

```
hours = [1, 2, 3, 4, 5]
scores = [49, 56, 65, 71, 82]
w, b, eta = 5, 40, 0.02 # first guess, fixed start, step size

for step in range(6):
    steepness = 0
    for x, y in zip(hours, scores):
        miss = y - (w * x + b)
        steepness += -2 * x * miss / len(hours)
    w = w - eta * steepness # roll downhill
    print(step + 1, round(w, 3))
```

Output (step, w): 1 6.4 · 2 7.184 · 3 7.623 · 4 7.869 · 5 8.007 · 6 8.084

Where this lives in real AI

	Today's example	A modern AI model
Input	1 number (hours)	thousands to millions of numbers
Model	a straight line	a neural network
Knobs	1 (just w)	billions, even trillions
Learning	roll down a curve	roll downhill, all knobs at once

Same three ideas. Same loop.

The road ahead

Season	Topic	Powers idea
0	The Toolkit: functions, slope, exponents, logs	all three
1	Vectors and Linear Algebra	Data
2	Matrix Decompositions	Data
3	Calculus from Zero	Learning
4	Probability	Model
5	Optimization	Learning
6	Putting It All Together	all three

THE EQUATION OF THE DAY

$$\hat{y} = wx + b$$

A prediction is a slope times an input, plus a starting point.

NEXT TIME • EPISODE 0.2

Reading Math Out Loud

Greek letters, subscripts, and the big $\sum$ from today's error formula